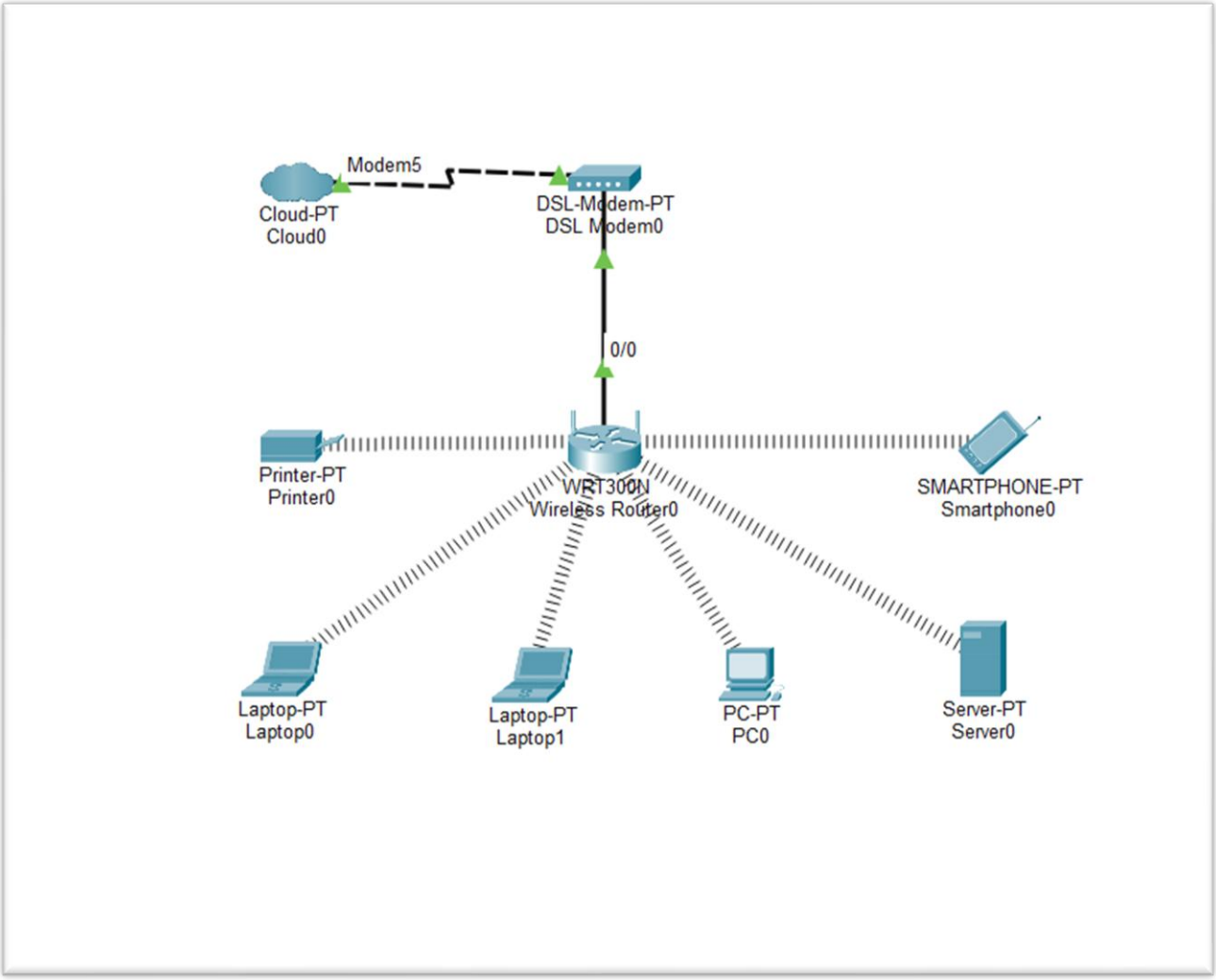
ASSIGNMENT--1

Q1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Answer:

This is a small office / home office network with one broadband (fibre) connection, one Wi-Fi router, two laptops, pc, server and one printer. Black solid lines are wired connections and dashed lines are wireless connections.



Connection details:

Wired Connections

1. **Cloud0 → DSL Modem0:** Connected with a phone line cable (dashed black line). The Modem5 port of the cloud is connected to the line port of the modem.
2. **DSL Modem0 → Wireless Router0 (WRT300N):** Connected with a copper straight-through Ethernet cable (solid black line). It plugs into the Internet port of the router.

Wireless Connections (Wi-Fi, no cable)

3. Wireless Router0 → Laptop0

4. Wireless Router0 → Laptop1
5. Wireless Router0 → Printer0
6. Wireless Router0 → Smartphone0
7. Wireless Router0 → PC0
8. Wireless Router0 → Server0

Q2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Answer:

The models below are common in Indian homes and offices. Prices are only approximate and change often, so please check the current price on Amazon or Flipkart before buying.

Component	Suggested brand / model	Approx. price (INR)	Qty
Wi-Fi router	TP-Link Archer C6 (AC1200) or TP-Link Archer AX23 (Wi-Fi 6)	2,000 to 3,500	1
Fibre modem (ONT)	Provided by the ISP (Jio, Airtel, ACT, etc.)	Free / rented	1
Laptops	HP 15s or Lenovo IdeaPad Slim 3 (Wi-Fi + Ethernet adapter if no LAN port)	35,000 to 50,000 each	2
Smart TV	Xiaomi / Redmi Smart TV 43 inch or Samsung Crystal 4K	22,000 to 30,000	1
Network printer	Brother DCP-L2541DW (wired + Wi-Fi laser) or HP LaserJet Pro with LAN port	14,000 to 20,000	1
Ethernet cables	Cat6 patch cables, 2 m and 5 m (D-Link, Amkette, Belkin)	150 to 400 each	4
UPS for router (optional)	APC BX600C-IN or a small 12V router UPS	2,500 to 4,500	1
Switch (only if more wired ports needed)	TP-Link TL-SG105 (5-port Gigabit)	800 to 1,200	0 or 1
Wi-Fi extender (only if signal is weak)	TP-Link TL-WA850RE or RE315	1,200 to 2,500	0 or 1

Q3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Answer:

These steps are for a TP-Link router. Other brands have almost the same menus with slightly different names.

Connect the router WAN port to the ONT LAN port with an Ethernet cable, and plug in the router power adapter. Wait about 2 minutes for it to start.

Connect one laptop to the router with an Ethernet cable (or to its default Wi-Fi name printed on the router sticker).

Open a browser and type the router address: 192.168.0.1 or tplinkwifi.net (it is written on the sticker at the bottom of the router).

Change the default admin password: on the first screen it asks to create a login password. If it does not, go to Advanced > System Tools > Administration and set a new strong password (at least 12 characters with letters, numbers and symbols). Do not keep "admin / admin".

Set up the internet (WAN): go to Advanced > Network > Internet. Choose the connection type given by the ISP (Dynamic IP for most fibre connections, or PPPoE and enter the username and password from the ISP). Save, and check that the status shows "Connected".

Set a unique SSID: go to Wireless settings. Change the network name from the default (like TP-Link_XXXX) to something unique, for example Ratan_Office_2.4G and Ratan_Office_5G. Do not put personal details like address or phone number in the name.

Enable WPA2 security: in the same Wireless settings, set Security to WPA2-PSK (Personal) with AES encryption. Do not use WEP or an open network.
Set a strong Wi-Fi password (at least 12 characters, not the same as the admin password). Save the settings, and the Wi-Fi will restart.
Turn off WPS (it can be misused to guess the PIN), and enable the Guest Network if visitors need Wi-Fi.

Check DHCP: go to Network > DHCP Server. Make sure DHCP is on, set the router IP to 192.168.1.1, the pool 192.168.1.100 to 192.168.1.199, and the DNS to the router or 8.8.8.8.
Update the firmware from Advanced > System Tools > Firmware Upgrade, and then reboot the router.

Reconnect the laptops and the smart TV to the new SSID using the new password, and check that the internet works.

Q4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Answer:

Network: 192.168.1.0/24, subnet mask 255.255.255.0, default gateway 192.168.1.1 (router), DNS 192.168.1.1 and 8.8.8.8. The DHCP pool is 192.168.1.100 to 192.168.1.199. The static IPs for the printer and TV are kept outside this pool (below .100), so that DHCP never gives the same address to another device and avoids an IP conflict. The MAC addresses below are made up for this assignment.

Device	MAC address (made up)	IP address	Assignment	Connection
Router (gateway)	A4:2B:B0:11:22:33	192.168.1.1	Static	Wired (WAN) + Wi-Fi AP
Laptop 1	3C:52:82:6A:1B:C4	192.168.1.101	DHCP	Wireless
Laptop 2	50:9A:4C:7E:D2:15	192.168.1.102	DHCP	Wired
Printer	00:1B:A9:4F:8C:30	192.168.1.20	Static	Wired
Smart TV	F8:A2:D6:5B:91:E7	192.168.1.21	Static	Wireless

How the static IPs are set:
Printer: open the printer control panel (or its web page) > Network / TCP-IP settings > change from Auto (DHCP) to Manual. Enter IP 192.168.1.20, subnet mask 255.255.255.0, gateway 192.168.1.1, DNS 192.168.1.1.
Smart TV: Settings > Network > Wi-Fi > Advanced / IP settings > change from DHCP to Manual. Enter IP 192.168.1.21, subnet mask 255.255.255.0, gateway 192.168.1.1, DNS 8.8.8.8.

Laptops: leave the network adapter on "Obtain an IP address automatically" and "Obtain DNS server address automatically". The router DHCP server gives them an address from the pool. I can check it with the command ipconfig /all.
Optional: in the router I can also use DHCP Address Reservation (using the MAC address), so the router always gives the same IP to a device.

Q5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Answer:

First, I check whether the problem is only with this laptop or with the whole network. I look at whether the other laptop, the TV and phones are working. If all of them have no internet, the problem is with the router or the ISP. If only one laptop fails, the problem is with that laptop. Below are the common causes.

Cause 1: Wi-Fi is off, or the wrong password / SSID

Diagnose: Check if Airplane mode is on, if the Wi-Fi switch / function key is off, and if the laptop shows the correct network name. Look for a message like "Incorrect password" or "Can't connect to this network".
Fix: Turn on Wi-Fi and turn off Airplane mode. Click "Forget" on the network, connect again and enter the correct Wi-Fi password. If it still fails, restart the laptop, and as a test connect with an Ethernet cable.

Cause 2: The laptop is not getting a valid IP address (DHCP problem)

Diagnose: Open Command Prompt and run ipconfig. If the IP starts with 169.254.x.x (APIPA), the laptop did not get an address from the router. A correct address should look like 192.168.1.1xx.
Fix: Run ipconfig /release and then ipconfig /renew. Check that the adapter is set to "Obtain an IP address automatically". If it still fails, restart the router and check in the router that DHCP is on and the pool is not full. Also check that nobody has set a static IP that conflicts with the pool.

Cause 3: DNS problem (can ping an IP but websites do not open)

Diagnose: Run ping 8.8.8.8. If it replies but ping google.com fails ("could not find host"), then DNS is the problem. Also check the DNS server value in ipconfig /all.
Fix: Run ipconfig /flushdns. Then change the DNS manually to 8.8.8.8 and 8.8.4.4 (or 1.1.1.1) in the adapter settings, or set the router DNS to the same. Restart the browser and test again.

Cause 4: Router or ISP problem

Diagnose: Run ping 192.168.1.1 (the router). If this works but ping 8.8.8.8 fails, the laptop is fine and the problem is after the router. Check the router lights (the internet / WAN light may be red or off) and the ONT lights (a red LOS light means the fibre is cut or the signal is lost).
Fix: Check that the cable between the ONT and router is firmly plugged in. Restart the ONT and router (switch off for 30 seconds, ONT first and then the router). If the ONT still shows a red light, call the ISP customer care.

Cause 5: Faulty cable, weak signal, or driver problem

Diagnose: For a wired laptop, check that the LAN port light is on and try another cable or port. For Wi-Fi, check the signal bars. In Device Manager, check that the network adapter has no yellow warning mark.
Fix: Replace the damaged cable or use another router port. Move closer to the router, or use the 5 GHz band / an extender. Update or reinstall the network adapter driver, or run "Network reset" in Windows settings.

Cause 6: Security settings blocking the laptop

Diagnose: Check the router for MAC filtering, parental controls or an access-control list that may be blocking this laptop. Also check the laptop for a wrong proxy or VPN setting, and for a firewall or antivirus blocking the network.
Fix: Add the laptop MAC address to the allowed list (or turn off MAC filtering), turn off the proxy / VPN, and temporarily disable the firewall to test. Turn security back on after finding the cause.

Finally, I test by opening a website, and I note down the problem and the fix for future reference.